25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Matrices, Matrix Algebra & Inverses Practice Exercises

1 MCQ

Find $\|\mathbf{u}\|$ for $\mathbf{u} = (-9, -12)$.

A 21
B 14
C 15
D 225

2 MCQ

A 3×3 symmetric matrix Q_k has eigenvalues $\lambda_1 = 12$, $\lambda_2 = 3$, and $\lambda_3 = 14$. Determine the trace $\text{tr}(Q_k)$.

A 26
B 32
C 29
D 41

3 MCQ

A linear transformation $T : \mathrm{R}^2 \to \mathrm{R}^2$ is defined by $T(\mathbf{x}) = A\mathbf{x}$, where $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, find $T(\mathbf{v})$.

A $\begin{pmatrix} 4 \\ 7 \end{pmatrix}$
B $\begin{pmatrix} 3 \\ 7 \end{pmatrix}$
C $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$
D $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$

4 MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. If a point $P = (x, y)$ is transformed by A to $P' = (x', y')$, such that $\begin{pmatrix} x' \\ y' \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix}$, what is the image of the point $(3, 1)$ under this transformation?

A (7, 6)
B (6, 7)
C (8, 5)
D (5, 8)

5 MCQ

Consider two vectors $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$. What is the dot product $\mathbf{v}_1 \cdot \mathbf{v}_2$?

A 1
B 0
C -7
D 25

6 MCQ

Consider the following system of linear equations represented by the augmented matrix:

$$\left(\begin{array}{cc|c} 1 & 2 & 5 \\ 2 & -1 & 0 \end{array}\right)$$

. Use row reduction to transform the matrix into row-echelon form and determine the unique solution (x, y) for the system. What is the value of x?

A 1

B 2

C 3

D 4

7 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 5 \\ -12 \end{pmatrix}$. Calculate the magnitude of $\mathbf{v}$, denoted as $\|\mathbf{v}\|$.

A 13

B 17

C 7

D 169

8 MCQ

Consider the vector $\mathbf{v} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$. If this vector is plotted on a standard xy-plane, it represents a directed line segment from the origin to the point $(3, 4)$. What is the geometric interpretation of the length of this vector?

A The vector's direction in the xy-plane.

B The angle the vector makes with the positive x-axis.

C The distance of the point (3,4) from the origin.

D The sum of the components of the vector.

9 MCQ

A planar linear transformation B scales a by 21 and y by 14 with matrix $T = \begin{pmatrix} 21 & 0 \\ 0 & 14 \end{pmatrix}$. Determine the matrix trace $\text{tr}(T)$.

A 38

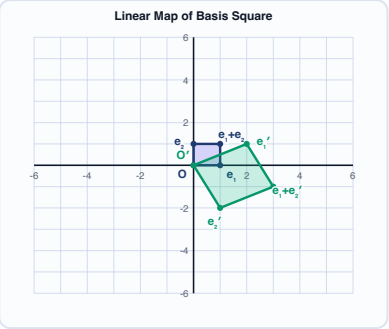
B 19

C 35

D 56

10 MCQ

Consider a linear transformation represented by the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}$. A point P is mapped to P' by this transformation. If the coordinates of P are $\begin{pmatrix} x \\ y \end{pmatrix}$ and the coordinates of P' are $\begin{pmatrix} x' \\ y' \end{pmatrix}$, then $\begin{pmatrix} x' \\ y' \end{pmatrix} = A \begin{pmatrix} x \\ y \end{pmatrix}$. Suppose a curve is defined by the equation $x'^2 + y'^2 = 5$. What is the equation of the original curve $x^2 + y^2 = k$ in terms of x and y that transforms into this circle? Find the value of k.



A 1

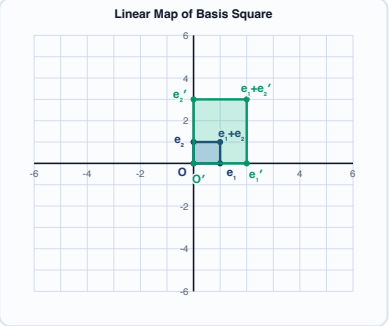
B 2

C 3

D 4

11 MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. If $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, what is $A\mathbf{v}$?



A $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

B $\begin{pmatrix} 2 \\ 6 \end{pmatrix}$

C $\begin{pmatrix} 1 \\ 6 \end{pmatrix}$

D $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$

12 MCQ

Consider the matrix $A = \begin{pmatrix} k & 2 \\ 1 & -3 \end{pmatrix}$. If the matrix A is invertible and its inverse A^{-1} has a determinant of $-\frac{1}{11}$, find the value of k.

A 3

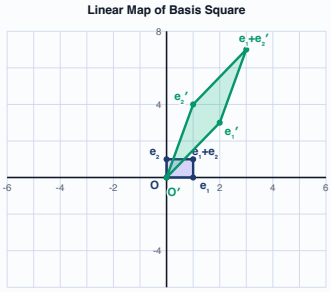
B 4

C 5

D 6

13 MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$. What is the matrix A^T?



A $\begin{pmatrix} 2 & 4 \\ 1 & 3 \end{pmatrix}$

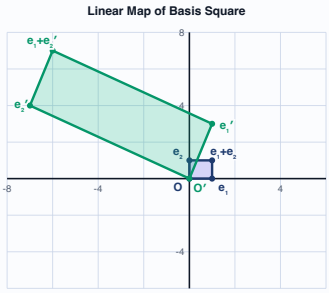
B $\begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix}$

C $\begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$

D $\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$

14 MCQ

Let A be a 2×2 matrix given by $A = \begin{pmatrix} 2 & k \\ 3 & -1 \end{pmatrix}$. If $A^2 = \begin{pmatrix} 1 & -7 \\ 3 & 4 \end{pmatrix}$, find the value of k.



A 2

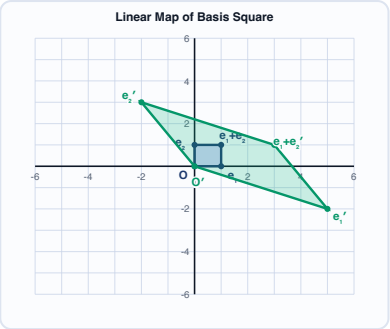
B 4

C 3

D 5

15 MCQ

For $A = \begin{bmatrix} 5 & -2 \\ -2 & 3 \end{bmatrix}$, find entry $(-3A)_{12}$.



A -2

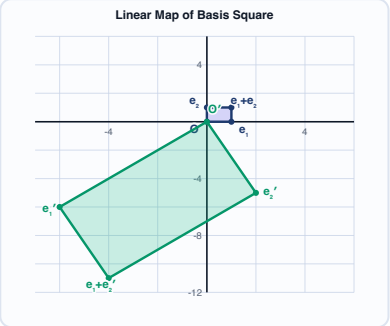
B 6

C -6

D 7

16 MCQ

For $A = \begin{bmatrix} -6 & 2 \\ -6 & -5 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -2 \\ 4 & 5 \end{bmatrix}$, find entry $(A + B)_{12}$.



A 2

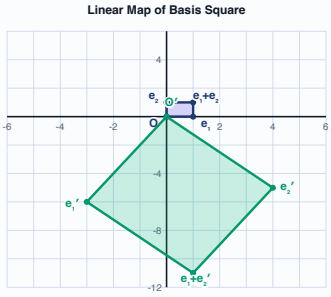
B 0

C -2

D 1

17 MCQ

For $A = \begin{bmatrix} -3 & 4 \\ -6 & -5 \end{bmatrix}$, $B = \begin{bmatrix} 6 & -1 \\ 2 & -5 \end{bmatrix}$, find entry $(A+B)_{22}$.



A -5

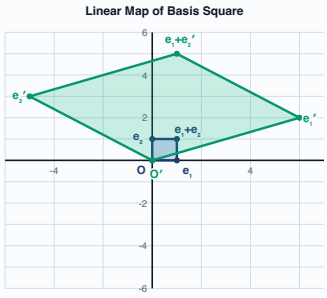
B -10

C -9

D -12

18 MCQ

For $A = \begin{bmatrix} 6 & -5 \\ 2 & 3 \end{bmatrix}$, find entry $(-3A)_{22}$.



A 3

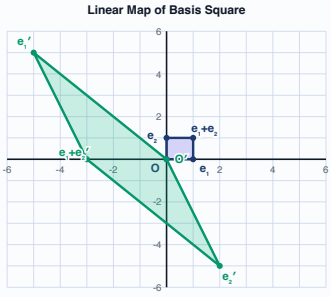
B 9

C -9

D -8

19 MCQ

For $A = \begin{bmatrix} -5 & 2 \\ 5 & -5 \end{bmatrix}$, find entry $(2A)_{22}$.



A -10

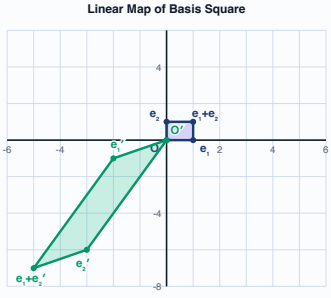
B -5

C 10

D -9

20 MCQ

For $A = \begin{bmatrix} -2 & -3 \\ -1 & -6 \end{bmatrix}$, find entry $(-3A)_{22}$.



A 19

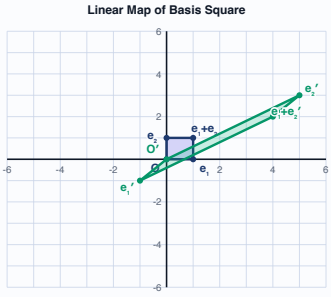
B -6

C -18

D 18

21 MCQ

For $A = \begin{bmatrix} -1 & 5 \\ -1 & 3 \end{bmatrix}$, find entry $(-4A)_{21}$.



A 4

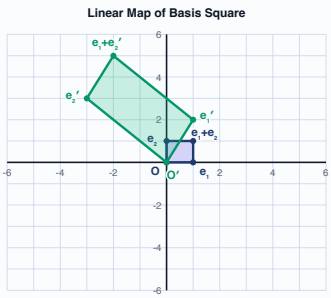
B 5

C -1

D -4

22 MCQ

For $A = \begin{bmatrix} 1 & -3 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 2 \\ 6 & -1 \end{bmatrix}$, find entry $(A + B)_{21}$.



A 2

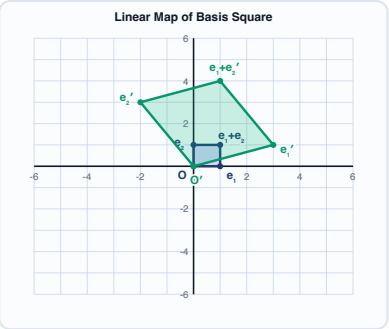
B 8

C 6

D 9

23 MCQ

For $A = \begin{bmatrix} 3 & -2 \\ 1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} -3 & -6 \\ -6 & 4 \end{bmatrix}$, find entry $(A+B)_{11}$.



A 3

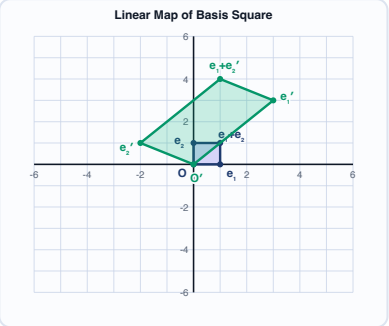
B 0

C -3

D 1

24 MCQ

For $A = \begin{bmatrix} 3 & -2 \\ 3 & 1 \end{bmatrix}$, find entry $(-3A)_{12}$.



A -2

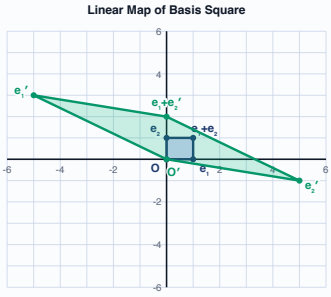
B -6

C 7

D 6

25 MCQ

For $A = \begin{bmatrix} -5 & 5 \\ 3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 6 & 1 \\ -3 & -6 \end{bmatrix}$, find entry $(A+B)_{22}$.



A -6

B -9

C -1

D -7